

마이크로파회로설계 최종 발표

2조 : 90도 하이브리드 커플러

조장 : 김용모

조원 : 김형용 김언석 김현진 한건희

목차

- 01. 설계목표
- 02. 3단 hybrid coupler 이론
- 03. RF에서 고 임피던스 구간 설계
- 04. 3단 hybrid coupler TLIN
- 05. 3단 hybrid coupler MLIN
- 06. 3단 hybrid coupler EM
- 07. 결과 및 분석

설계 목표

이상적인 하이브리드 커플러

$$S_{11} \rightarrow -\infty \text{dB}$$

$$S_{41} \rightarrow -\infty \text{dB}$$

$$S_{21} = -1.25 \text{dB}$$

$$S_{31} = -6.02 \text{dB}$$

S_{21} , S_{31} 간 90Deg 위상차이

대역폭 정의(목표값)

$$S_{11} < -20 \text{dB}$$

$$S_{41} < -20 \text{dB}$$

$$-1.71 \text{dB} < S_{21} < -0.84 \text{dB} \quad (10\% \text{오차})$$

$$-6.48 \text{dB} < S_{31} < -5.61 \text{dB} \quad (10\% \text{오차})$$

$$f_c \text{에서의 Phase error} < 0.1 \text{ Deg}$$

3단 hybrid coupler 이론

1. 직렬 임피던스 Z

$$M = \begin{bmatrix} 1 & Z \\ 0 & 1 \end{bmatrix}$$

2. 병렬 어드미턴스 Y

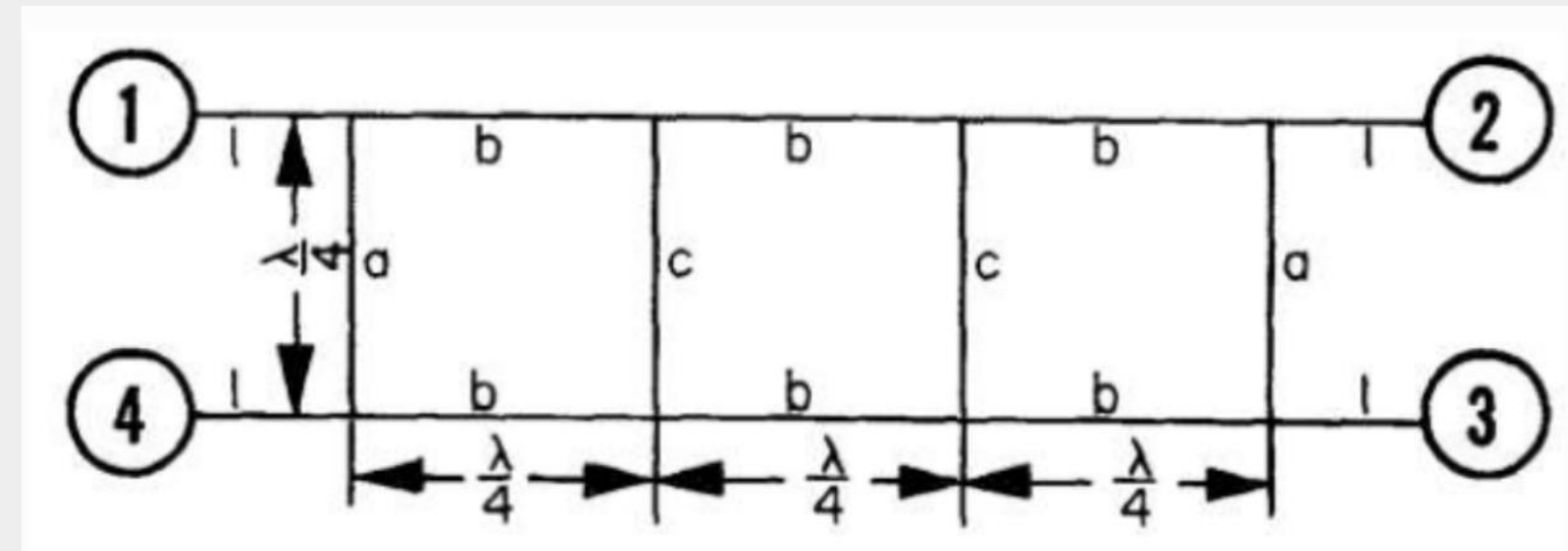
$$M = \begin{bmatrix} 1 & 0 \\ Y & 1 \end{bmatrix}$$

3. 무손실 전송선 (길이 l , 임피던스 z)

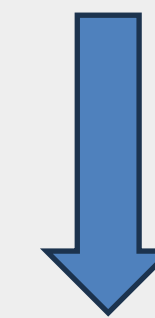
$$M = \begin{bmatrix} \cos(\beta l) & jz \sin(\beta l) \\ j\frac{1}{z} \sin(\beta l) & \cos(\beta l) \end{bmatrix}$$

- (여기서 $\beta l = \frac{2\pi l}{\lambda}$)

각 선로를 대칭으로 a, b, c로 정의
ABCD 행렬로 정리



$$M_{+\pm} = \begin{bmatrix} 1 & 0 \\ \pm ja & 1 \end{bmatrix} \begin{bmatrix} 0 & \frac{j}{b} \\ jb & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \pm jc & 1 \end{bmatrix} \begin{bmatrix} 0 & \frac{j}{b} \\ jb & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \pm jc & 1 \end{bmatrix} \begin{bmatrix} 0 & \frac{j}{b} \\ jb & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \pm ja & 1 \end{bmatrix}$$



$$M_{+\pm} = \begin{bmatrix} \pm \frac{1}{b} \left(a + c - \frac{ac^2}{b^2} \right) & \frac{j}{b} \left(\frac{c^2}{b^2} - 1 \right) \\ j \left(\frac{a^2 + 2ac}{b} - \frac{a^2 c^2}{b^3} - b \right) & \pm \frac{1}{b} \left(a + c - \frac{ac^2}{b^2} \right) \end{bmatrix}$$

3단 hybrid coupler 이론

$$M_{+\pm} = \begin{bmatrix} \pm \frac{1}{b} \left(a + c - \frac{ac^2}{b^2} \right) & \frac{j}{b} \left(\frac{c^2}{b^2} - 1 \right) \\ j \left(\frac{a^2 + 2ac}{b} - \frac{a^2 c^2}{b^3} - b \right) & \pm \frac{1}{b} \left(a + c - \frac{ac^2}{b^2} \right) \end{bmatrix}$$

정합 조건에 대한 식과
전력분배에 대한 식으로 정리.

1. 매칭 조건 (반사 없음):

$$c^2(1 + a^2) = b^2(1 + a^2 + 2ac - b^2)$$

2. 분배 조건 (1:3 분배):

$$(b^2(a + c) - ac^2)^2 + (c^2 - b^2)^2 = 2b^3(b^2(a + c) - ac^2)$$

$$\Gamma = 0$$

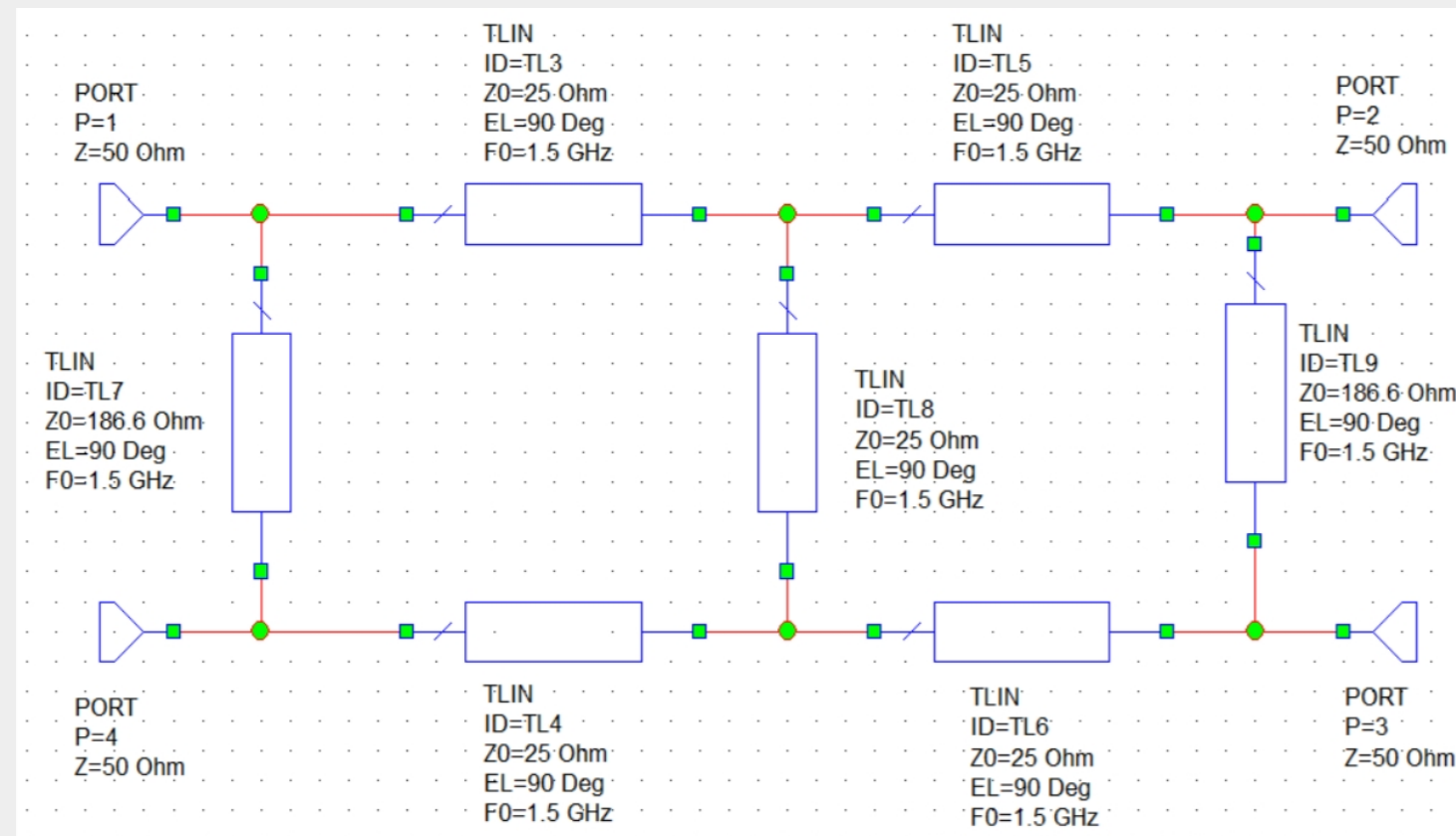
$$\Gamma = \frac{A + B - C - D}{A + B + C + D} = 0, \quad A + B - C - D = 0$$

$$\text{coupling} = 20 \log \frac{1}{|A_3|} = 6.02$$

$$T = \frac{2}{A + B + C + D}$$

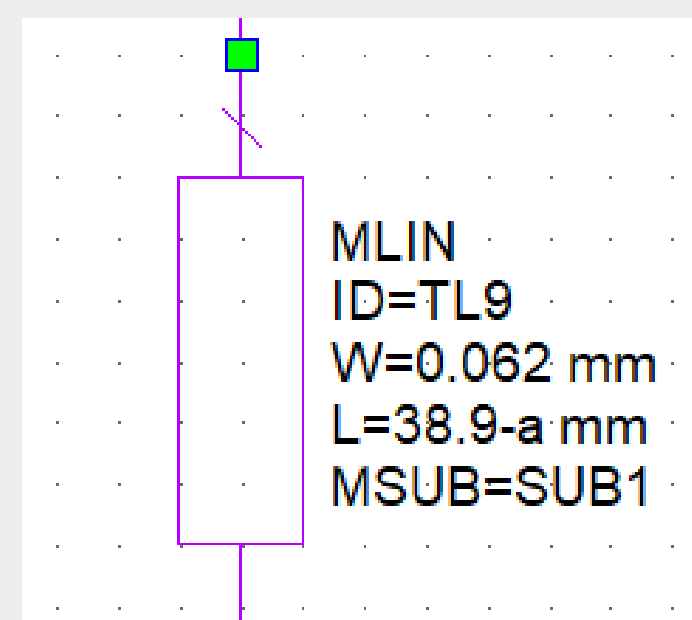
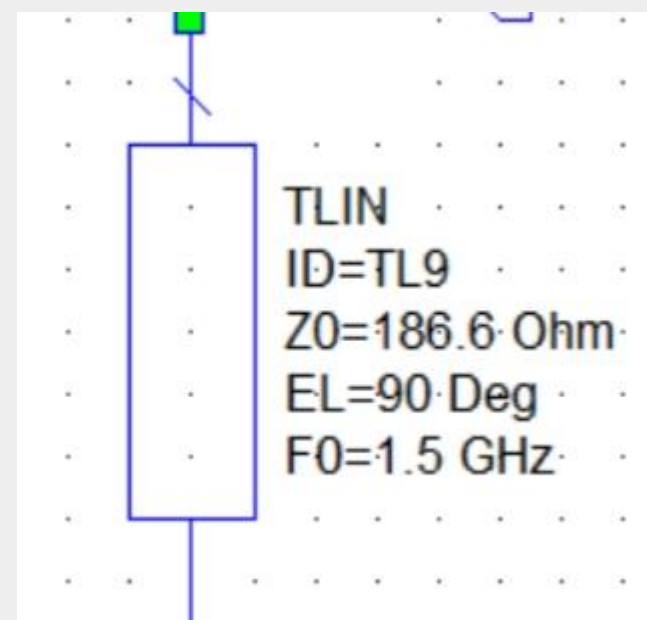
$$A_3 = S_{31} = \frac{1}{2}T_{++} - \frac{1}{2}T_{\pm}$$

RF에서 고 임피던스 구간 설계



높은 임피던스를 가지게 되면 선로가 얇아진다.

Ex) 186.6ohm -> 0.062mm



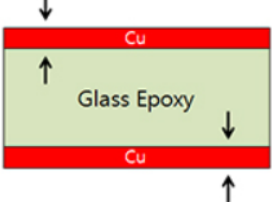
RF에서 고 임피던스 구간 설계

Traces			
Features	Capability	Description	Patterns
Min. track width and spacing (1 oz)	0.10 / 0.10 mm (4 / 4 mil)	1- and 2-layer: 0.10 / 0.10 mm (4 / 4 mil) Multilayer: 0.09 / 0.09 mm (3.5 / 3.5 mil). 3 mil is acceptable in BGA fan-outs.	
Min. track width and spacing (2 oz)	0.16 / 0.16 mm (6.5 / 6.5 mil)	2-layer: 0.16 / 0.16 mm (6.5 / 6.5 mil) Multilayer: 0.16 / 0.20 mm (6.5 / 8 mil)	
Min. track width and spacing (2.5 oz)	2 layer: 0.2/0.2mm(8mil/8mil)		
Min. track width and spacing (3.5 oz)	2 layer: 0.25/0.25mm(10mil/10mil)		
Min. track width and spacing (4.5 oz)	2 layer: 0.3/0.3mm(12mil/12mil)		

JLC PCB – 최소 0.1mm 요구

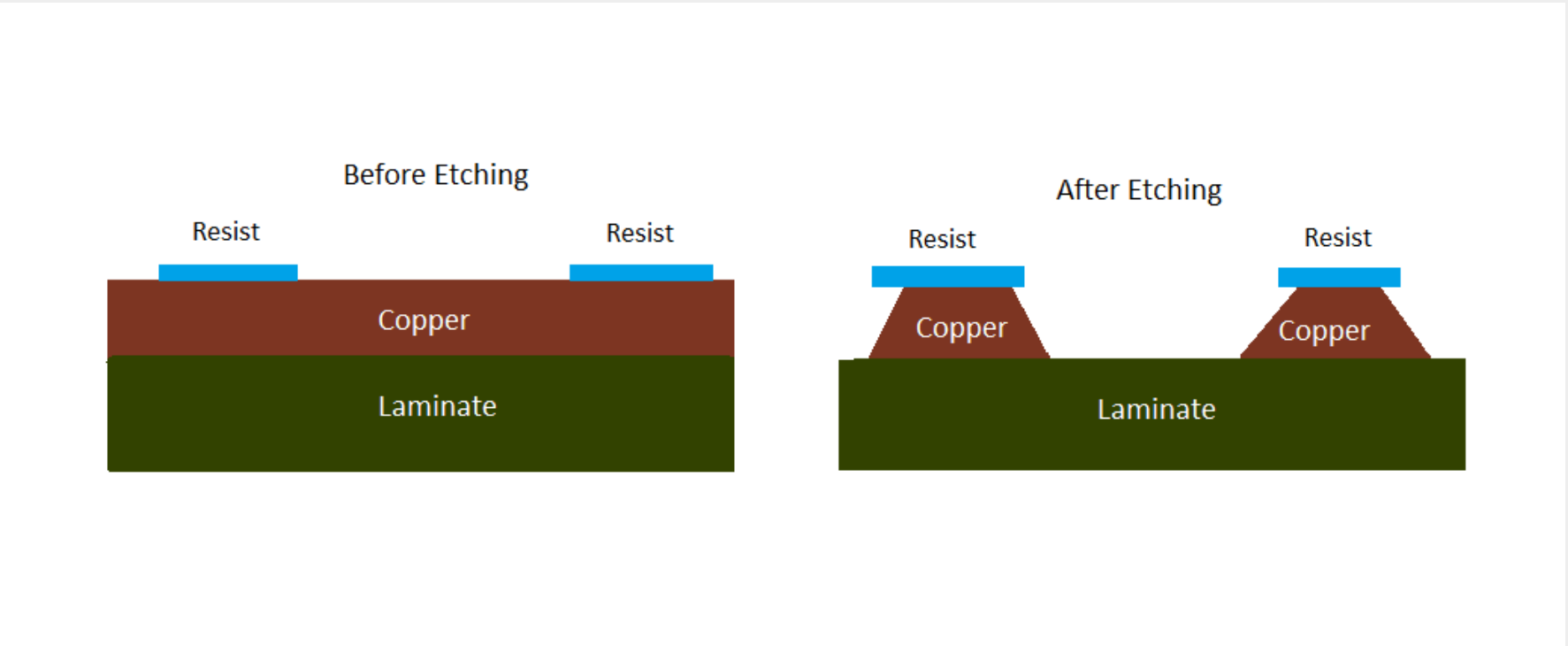
한샘 디지텍 – 최소 0.1mm 요구

동판 에칭중 발생하는 오차에 대비해
충분히 여유 있는 두께를 확보할 필요가 있음.

Pattern Line&Space			
Base Copper	Line width & space		
	Base Copper + Plating	Base Copper(Non-Plating) (1Layer & Inner Layer)	Base Copper + Plating(2회) (Filled Via 공법)
1/3oz(0.011mm)	0.100mm (4mils)	0.100mm (4mils)	0.125mm (5mils)
1/2oz(0.018mm)	0.100mm (4mils)	0.100mm (4mils)	0.150mm (6mils)
1oz(0.035mm)	0.125mm (5mils)	0.100mm (4mils)	0.200mm (8mils)
2oz(0.070mm)	0.150mm (6mils)	0.150mm (6mils)	0.250mm (10mils)
3oz(0.105mm)	0.200mm (8mils)	0.200mm (8mils)	N/A
			

→ Line&Space 0.075mm(3mils) 국부적 사용 가능

→ Copper foil 1/3oz, 3oz는 4Layer 이상 “Outer layer” 적용 가능

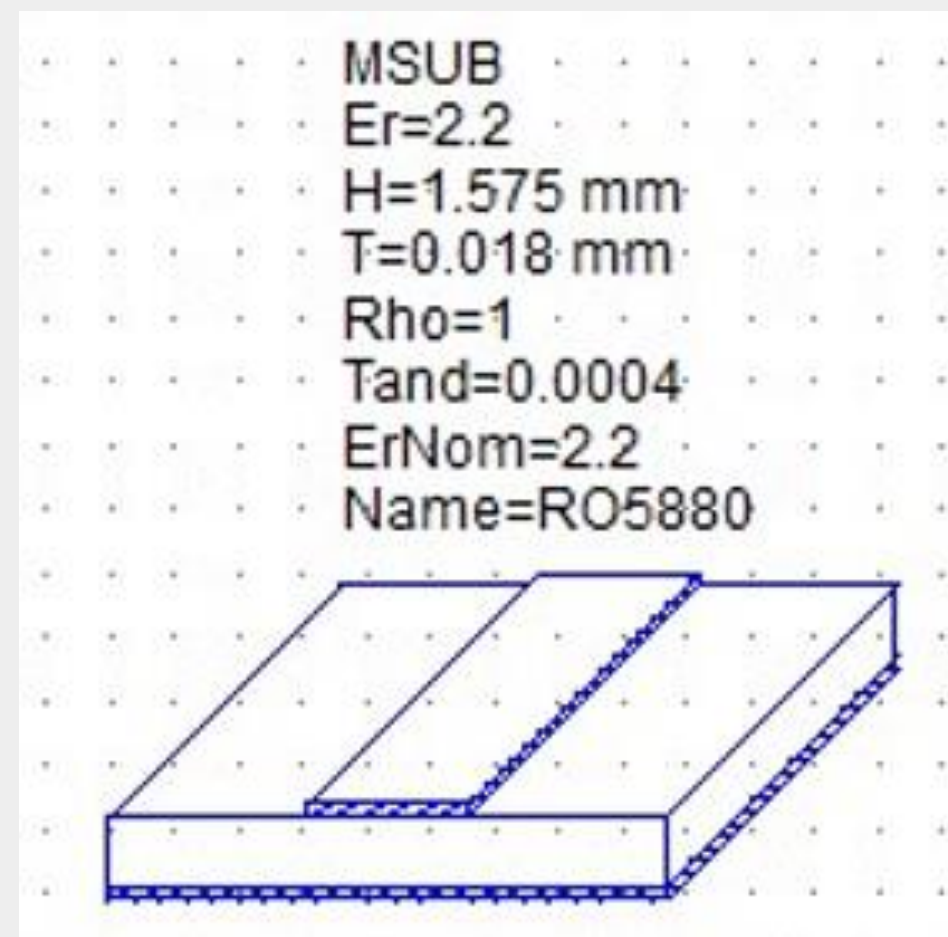


RF에서 고 임피던스 구간 설계

해결방안

1. 비 유전률이 낮은 기판 사용
2. 두꺼운 기판 사용
3. 낮은 특성 임피던스로 설계 후 $\lambda/4$ 임피던스 변환기로 정합
4. 요구조건 허용범위 내에서 오차 안정성 확보

$$Z_0 \approx \sqrt{\frac{L}{C}}$$



PROPERTY	TYPICAL VALUES	
	RT/duroid 5870	RT/duroid 5880
^[1] Dielectric Constant, ϵ_r Process	2.33 2.33 $\pm$ 0.02 spec.	<u>2.20</u> 2.20 $\pm$ 0.02 spec.
^[4] Dielectric Constant, ϵ_r Design	2.33	2.20
Dissipation Factor, $\tan \delta$	0.0005 0.0012	0.0004 0.0009
Thermal Coefficient of ϵ_r	-115	-125
Volume Resistivity	2 X 10 ⁷	2 X 10 ⁷
Surface Resistivity	2 X 10 ⁷	3 X 10 ⁷

Standard Thicknesses
0.005" (0.127mm) +/- 0.0005"
0.010" (0.252mm) +/- 0.0007"
0.020" (0.508mm) +/- 0.0015"
0.031" (0.787mm) +/- 0.0020"
<u>0.062" (1.575mm) +/- 0.0030"</u>
*Additional thicknesses available from 0.0035" - 0.375" in varying increments

Standard Claddings	
<u>Electrodeposited Copper Foil</u> ½ oz. (18µm) HH/HH 1 oz. (35µm) H1/H1	<u>Rolled Copper Foil</u> ½ oz. (18µm) 5R/5R 1 oz. (35µm) 1R/1R

3단 hybrid coupler TLIN

1. 매칭 조건 (반사 없음):

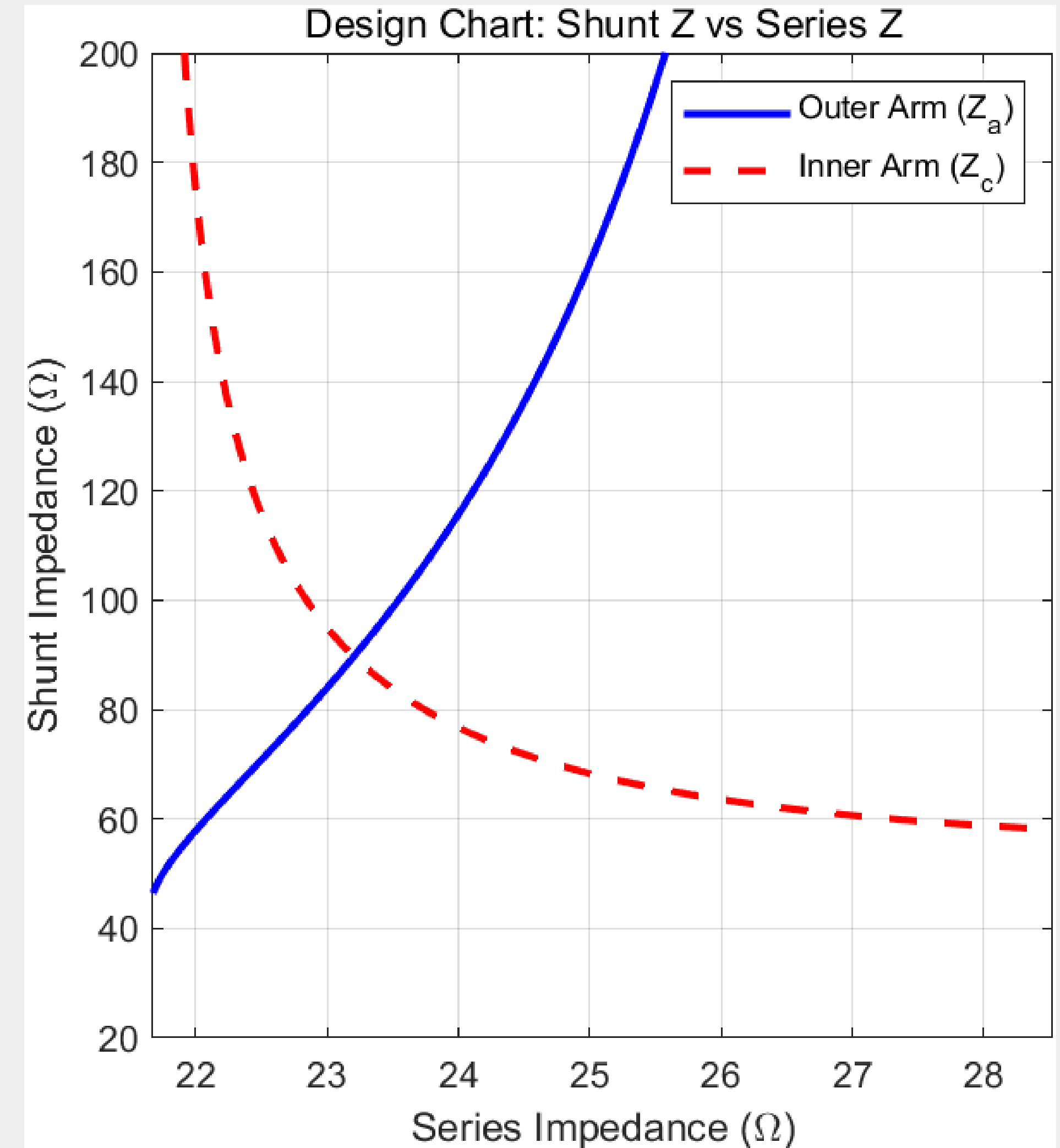
$$c^2(1 + a^2) = b^2(1 + a^2 + 2ac - b^2)$$

2. 분배 조건 (1:3 분배):

$$(b^2(a + c) - ac^2)^2 + (c^2 - b^2)^2 = 2b^3(b^2(a + c) - ac^2)$$

높은 a 임피던스

낮은 a 임피던스



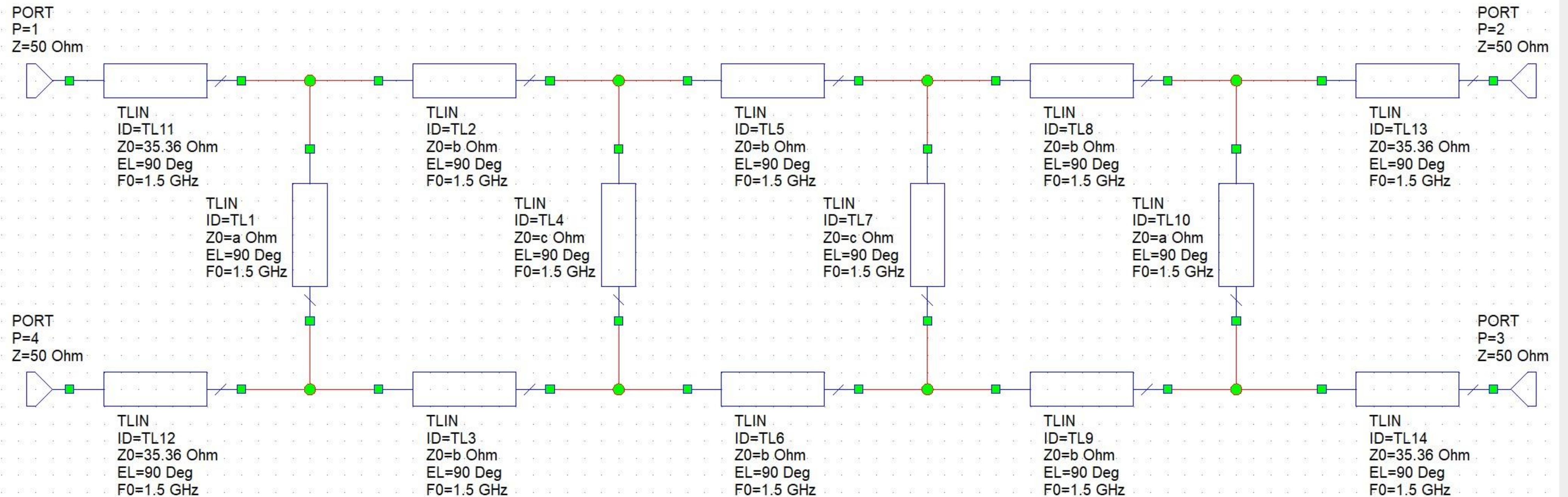
160ohm -> W : 0.36mm

100ohm -> W : 1.38mm >> 0.1mm

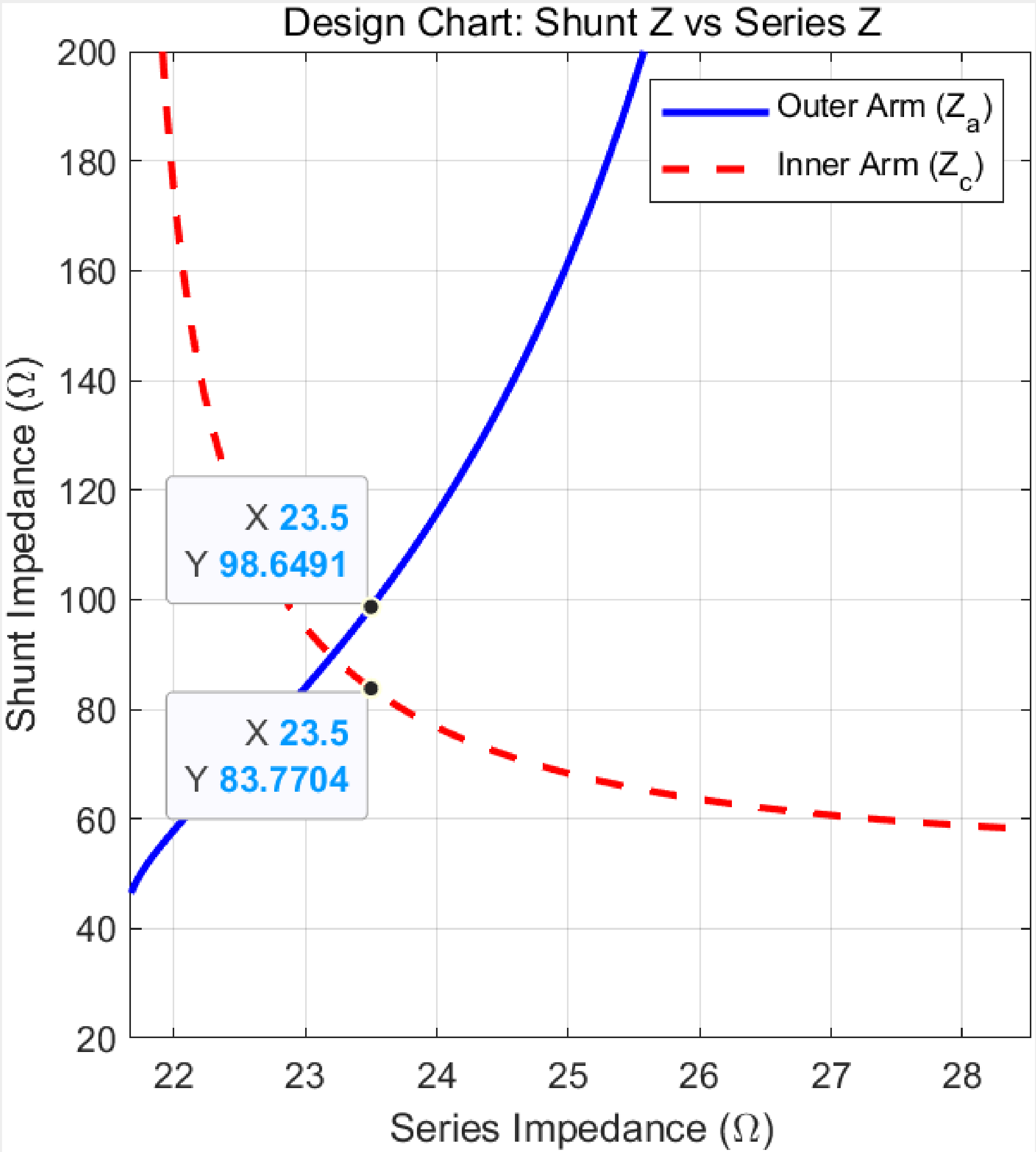
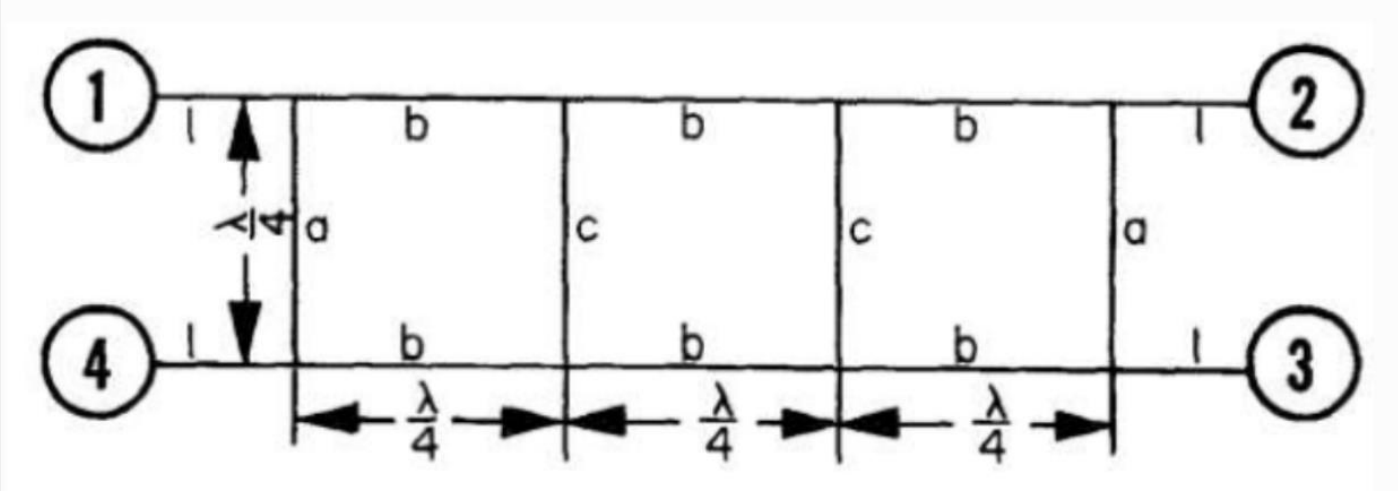
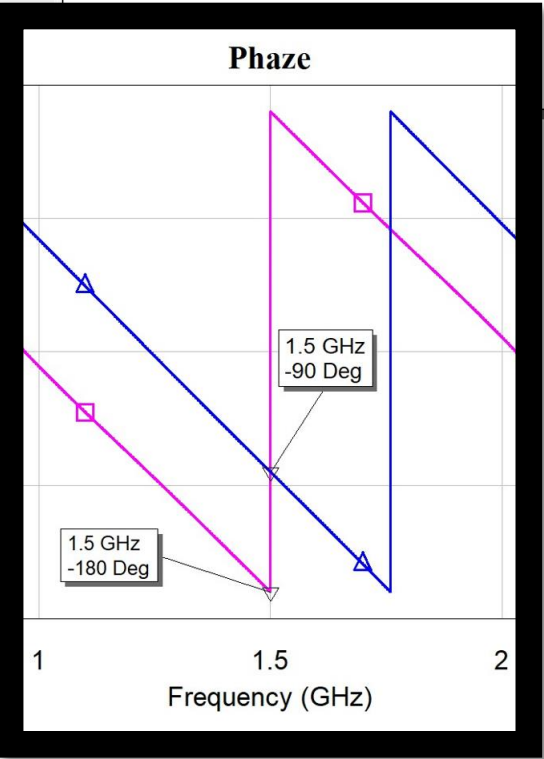
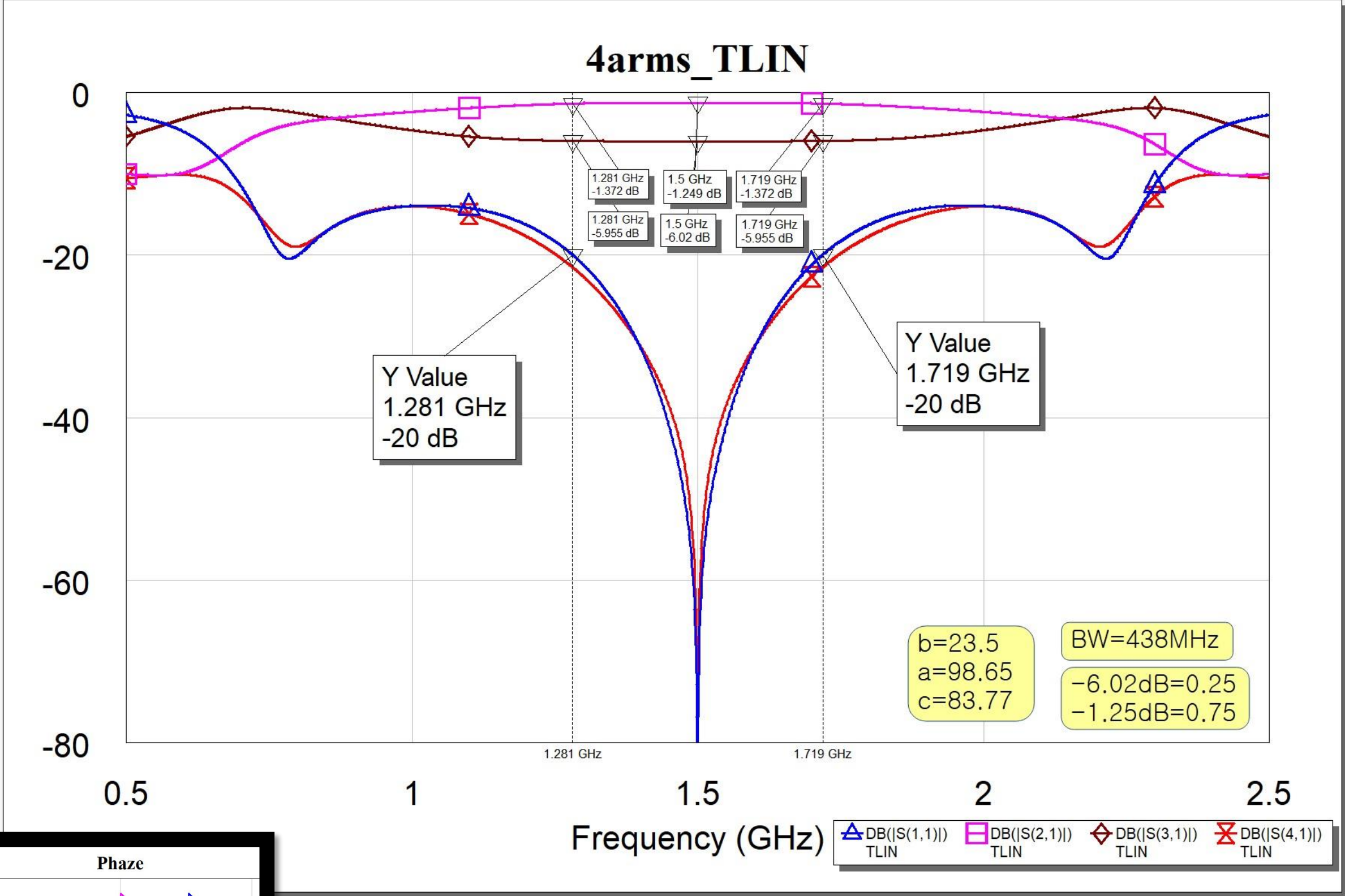
3단 hybrid coupler TLIN

$$b=23.5 \quad x = b/25 \quad Z0=25$$

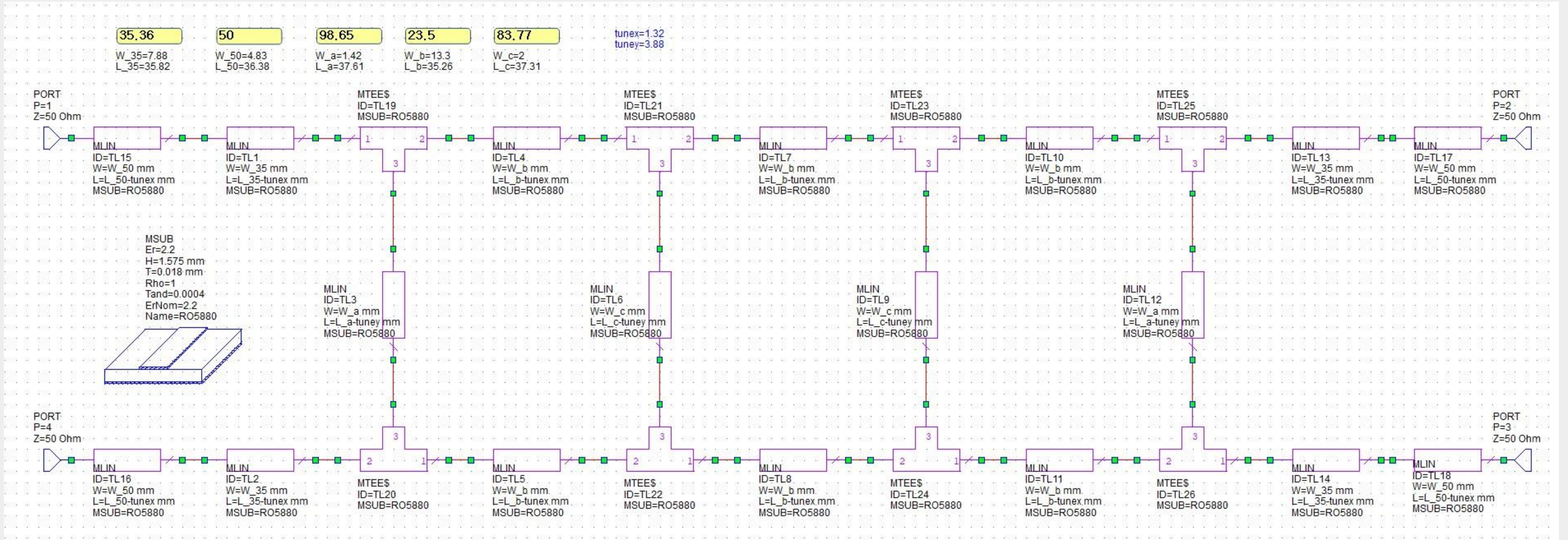
$$a=25 / ((14057.447093)*x^6 + (-86257.129986)*x^5 + (220292.642528)*x^4 + (-299740.694217)*x^3 + (229180.132590)*x^2 + (-93369.203322)*x + (15836.959614))$$
$$c=25 / ((-14384.458759)*x^6 + (88282.274322)*x^5 + (-225517.552177)*x^4 + (306930.885402)*x^3 + (-234747.865085)*x^2 + (95669.533621)*x + (-16232.450894))$$



3단 hybrid coupler TLIN



3단 hybrid coupler MLIN

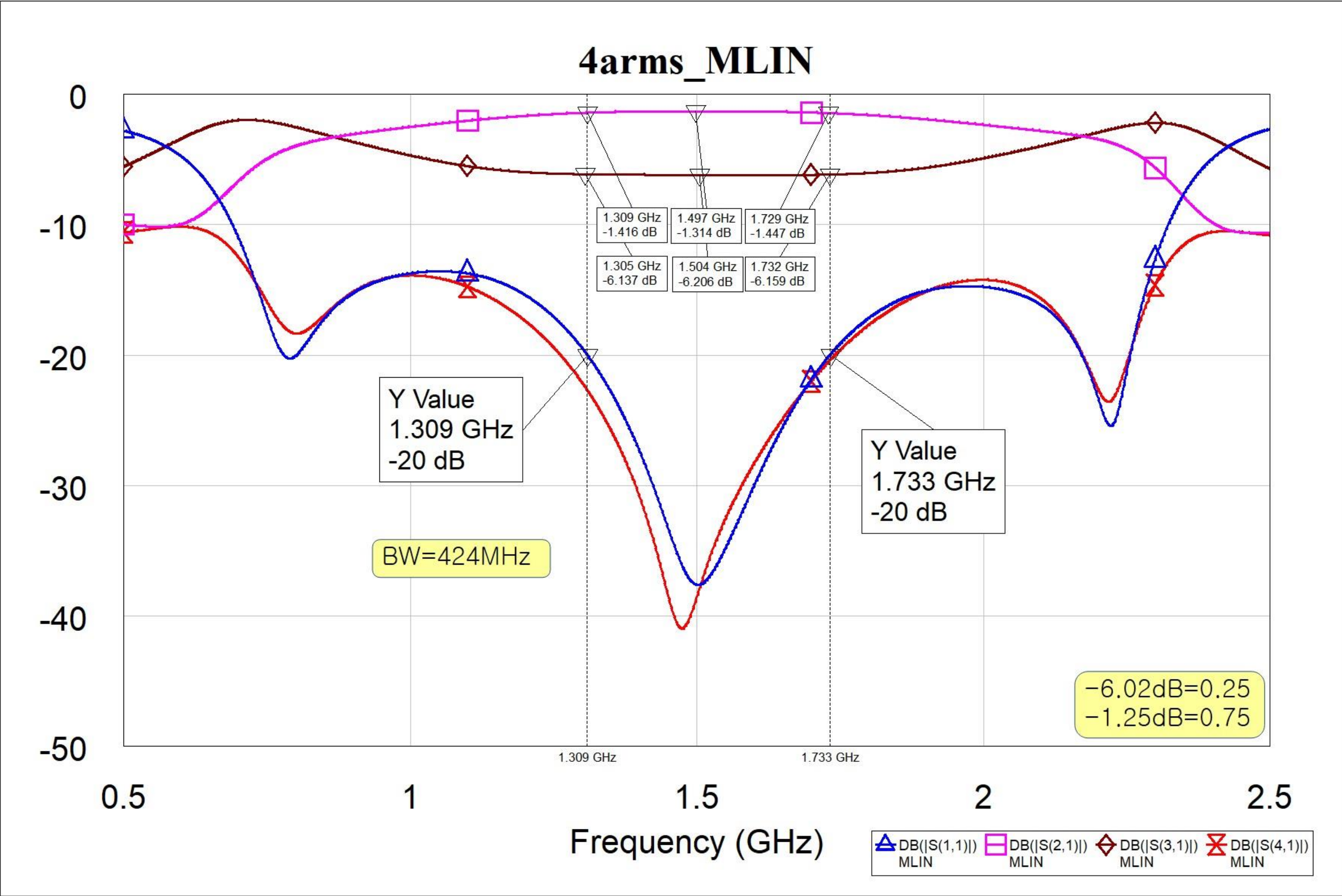


1/4파장 변환기

$$Z_T = \sqrt{Z_{in} \cdot Z_{load}}$$

$$Z_T = \sqrt{25 \times 50} = \sqrt{1250} \approx 35.355 \Omega$$

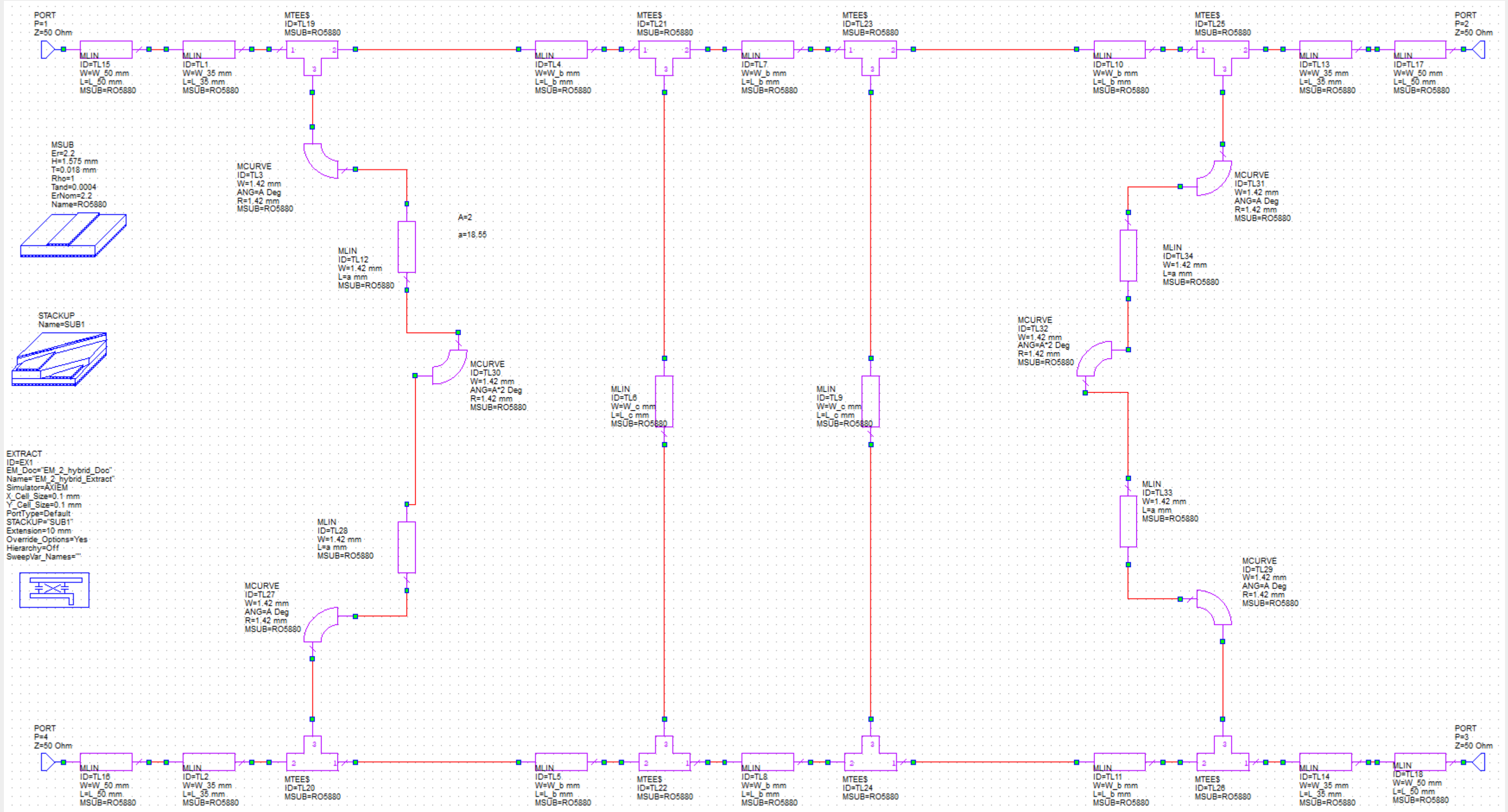
3단 hybrid coupler MLIN



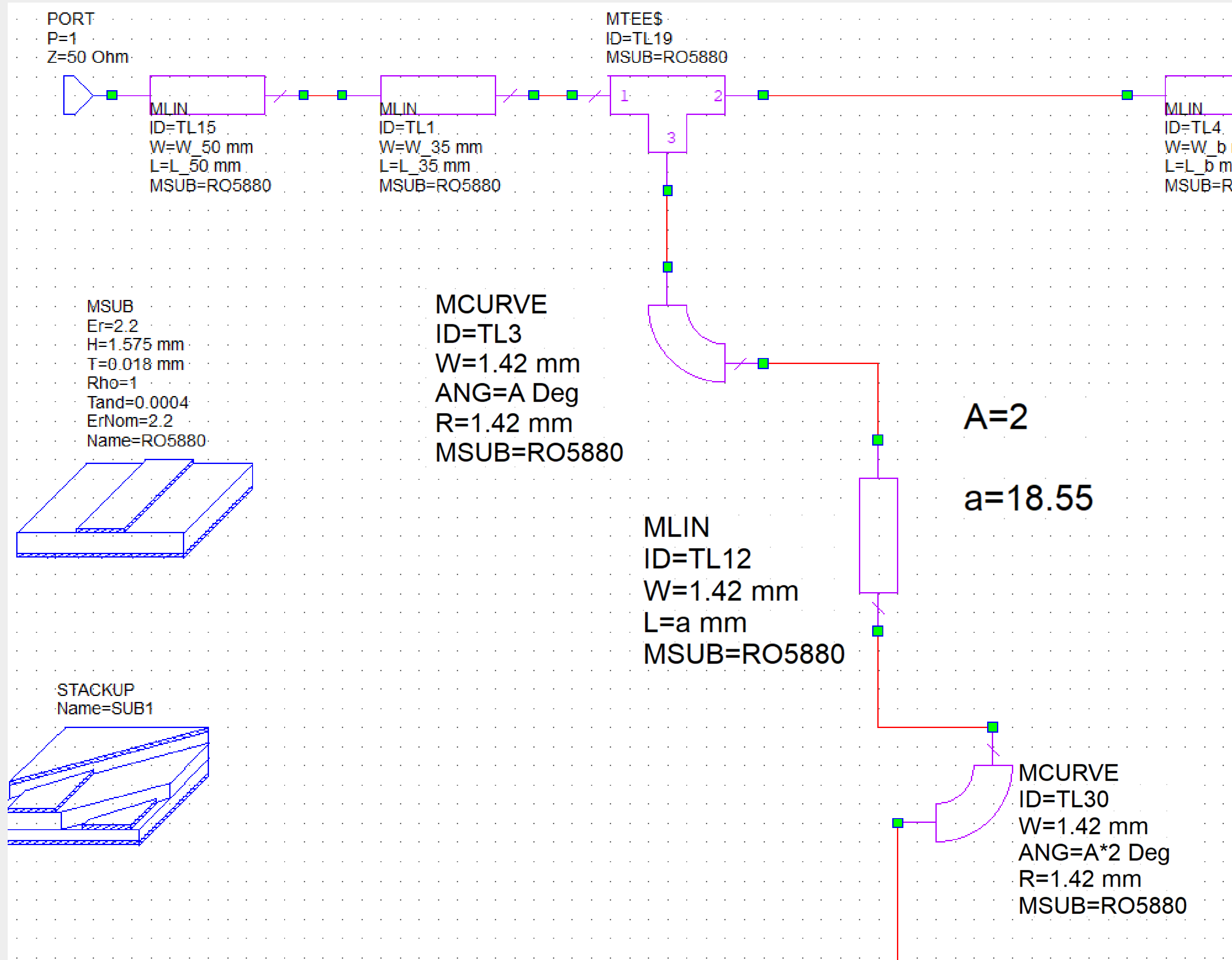
89.95 Deg < 0.1 Deg error

3단 hybrid coupler EM

35.35	50	98.65	23.5	83.77
W_35=7.88 L_35=34.5	W_50=4.83 L_50=35.06	W_a=1.42 L_a=37.61	W_b=13.3 L_b=33.94	W_c=2 L_c=37.31

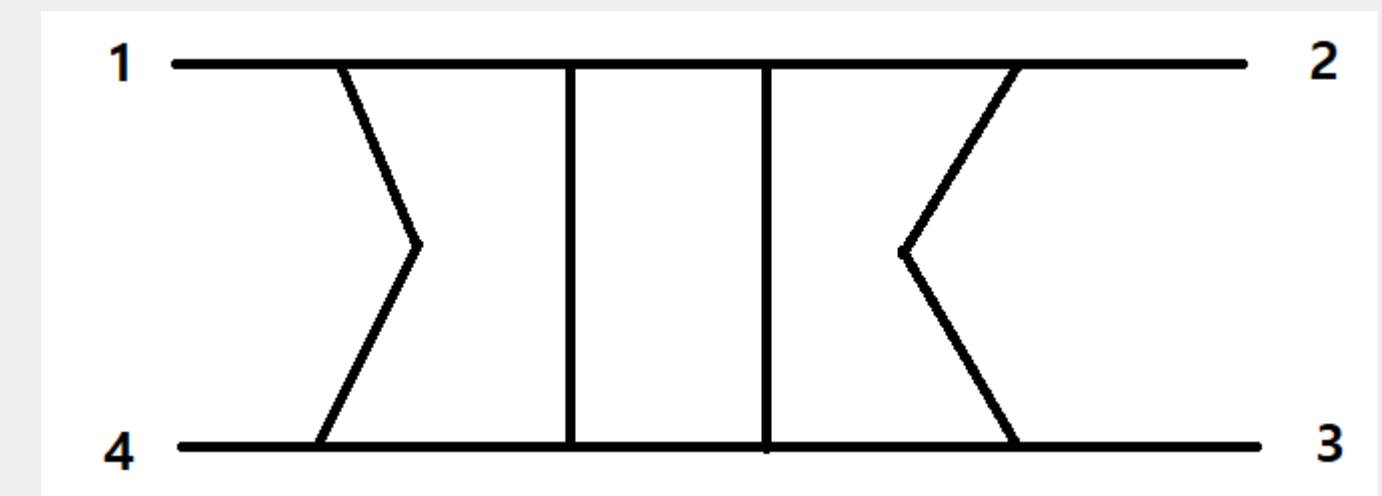


3단 hybrid coupler EM



98.65

$W_a = 1.42$
 $L_a = 37.61$

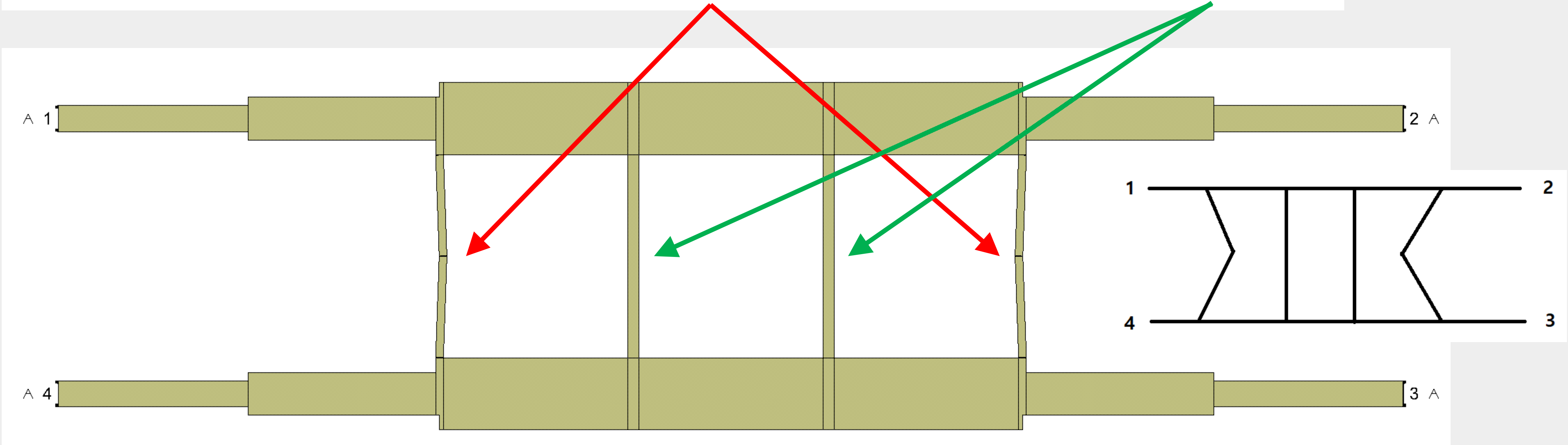


a, c 간 0.3mm 길이차이, 선로 두께보다 얇음

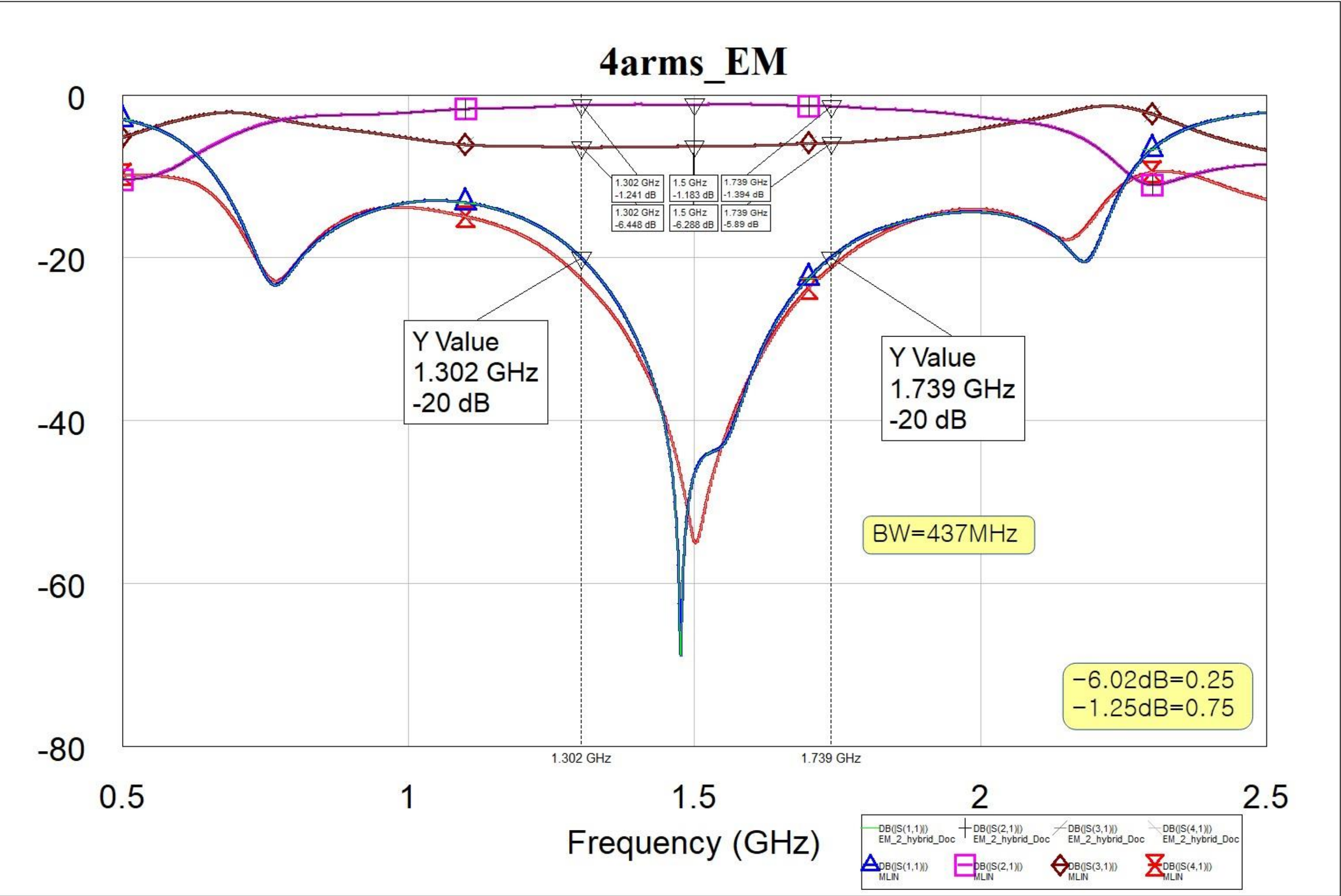
-> 90도의 MSUB 사용 불가

3단 hybrid coupler EM

35.35	50	98.65	23.5	83.77
$W_{35}=7.88$ $L_{35}=34.5$	$W_{50}=4.83$ $L_{50}=35.06$	$W_a=1.42$ $L_a=37.61$	$W_b=13.3$ $L_b=33.94$	$W_c=2$ $L_c=37.31$



3단 hybrid coupler EM



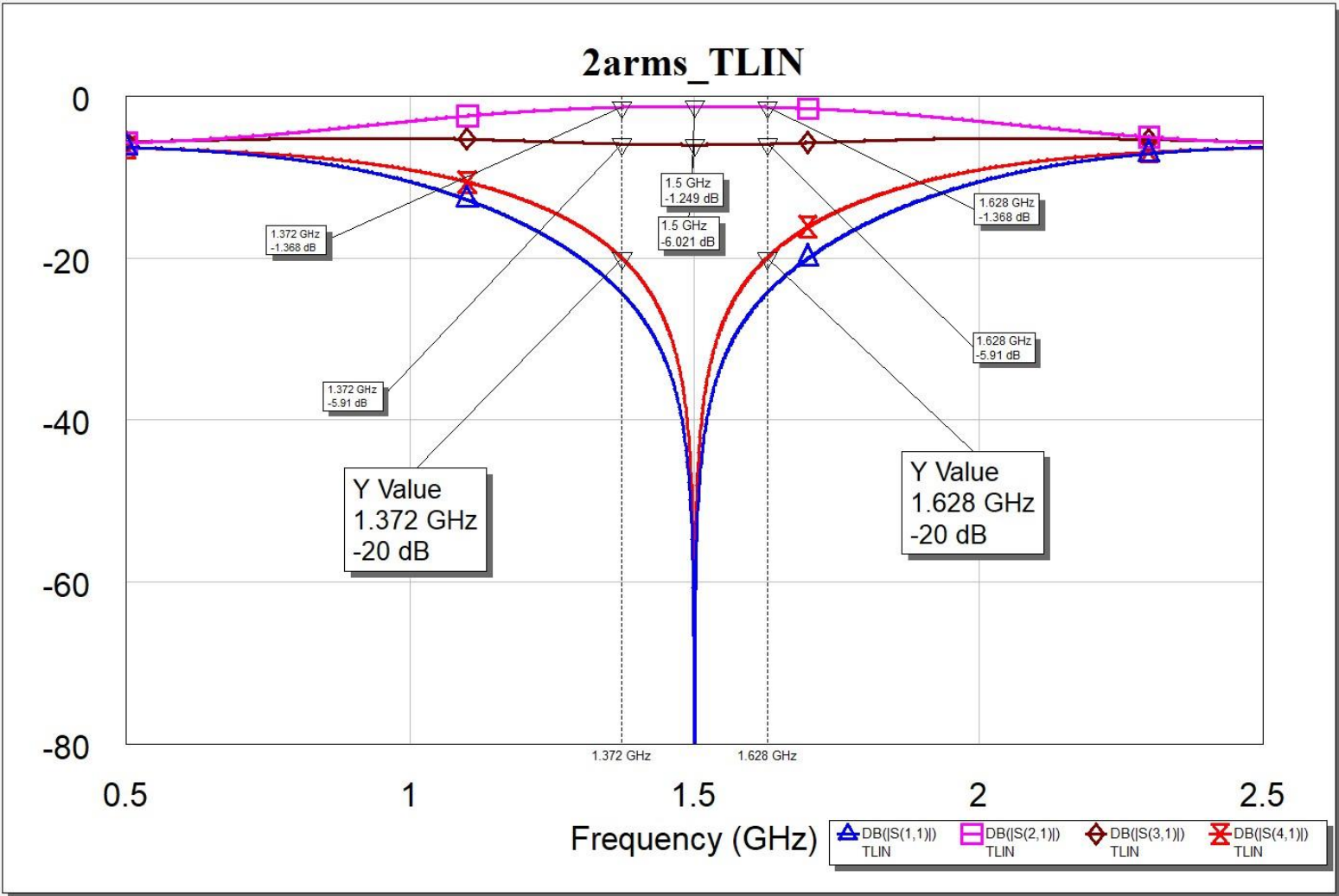
90.08 Deg < 0.1 Deg error

결과 및 분석

1, 2, 3단 커플러 대역, 특성 비교

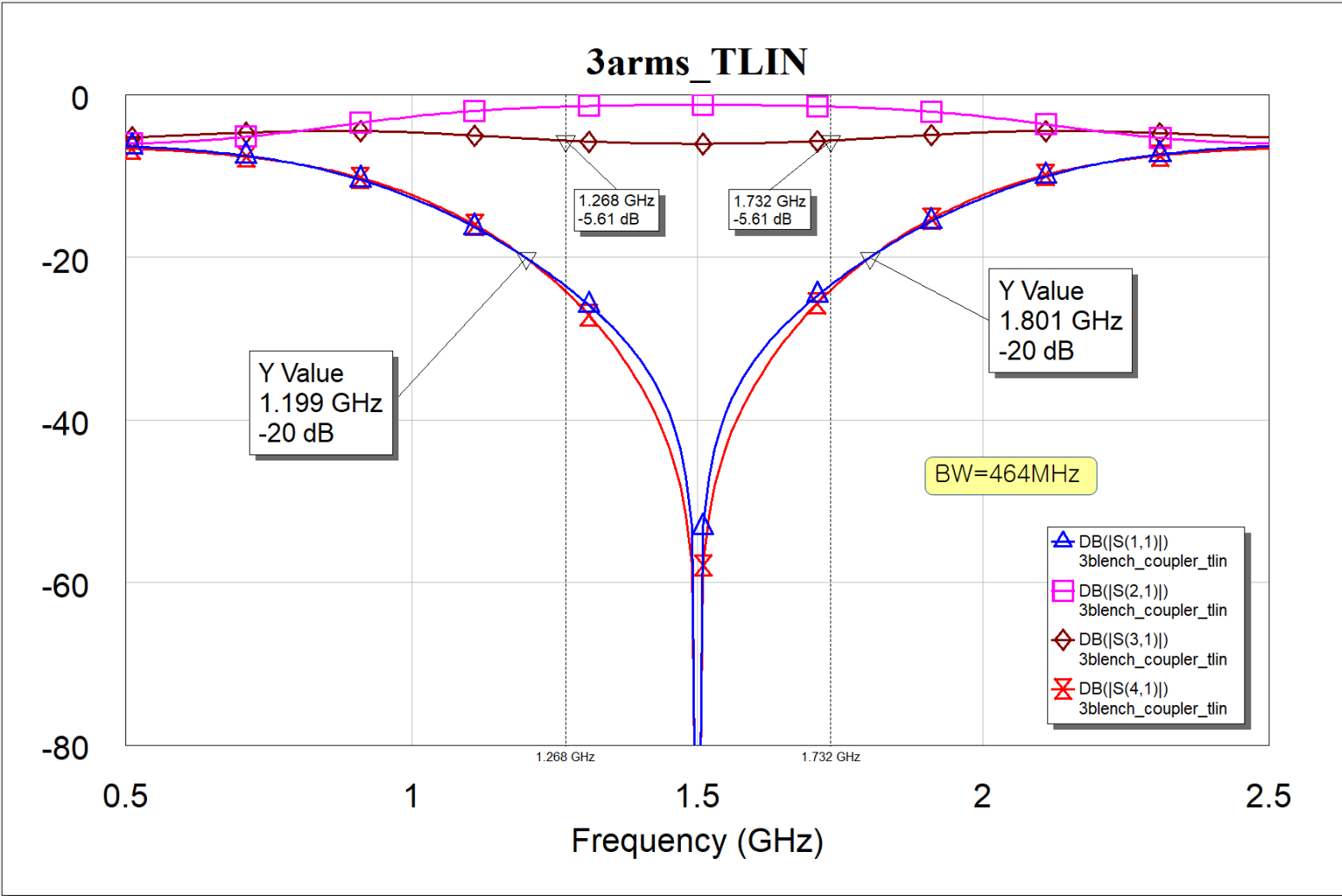
결과 및 분석

1단 TLIN



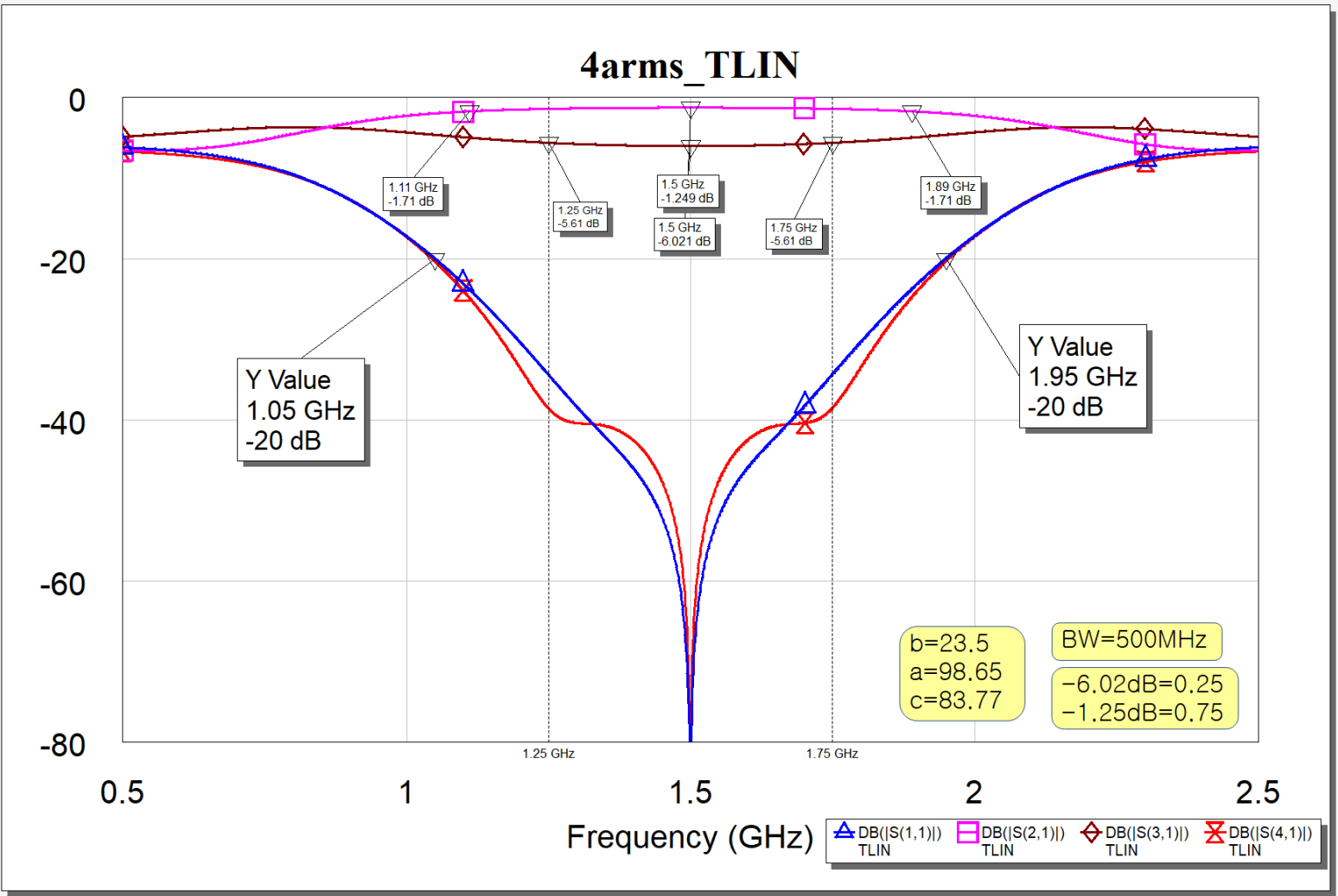
1단 대역폭 : 256MHz

2단 TLIN



2단 대역폭 : 464MHz

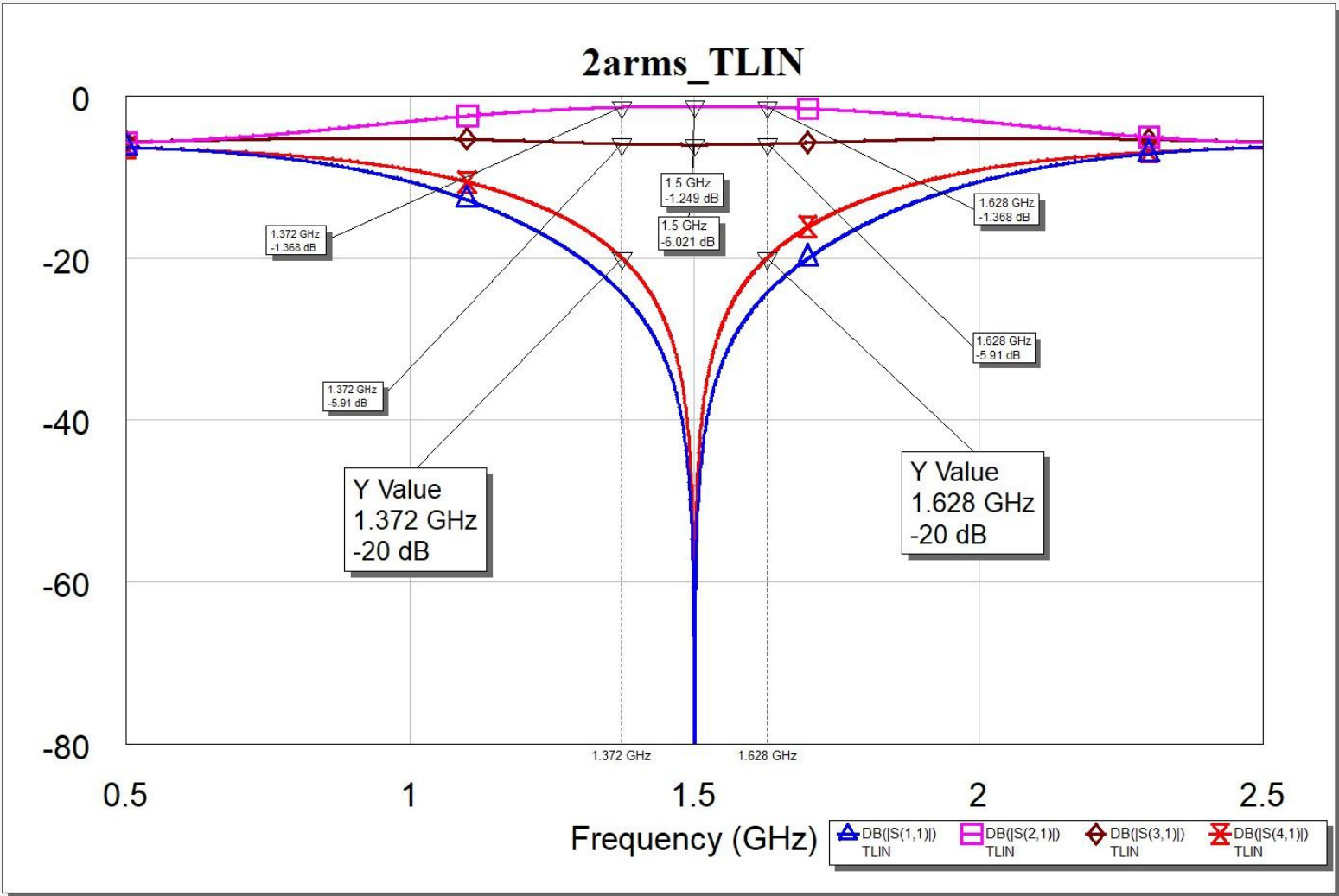
3단 25ohm TLIN



3단 대역폭 : 500MHz

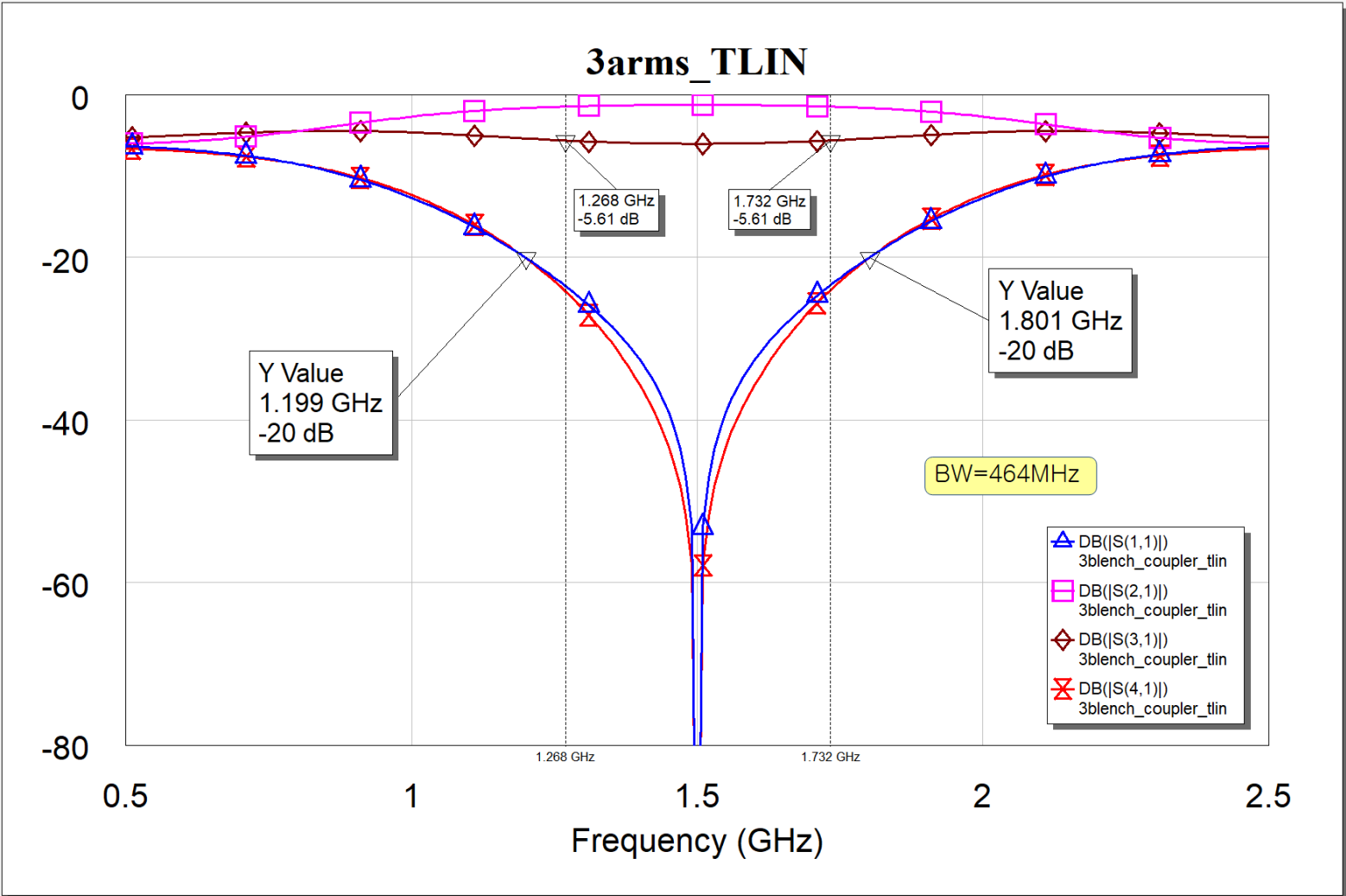
결과 및 분석

1단 TLIN



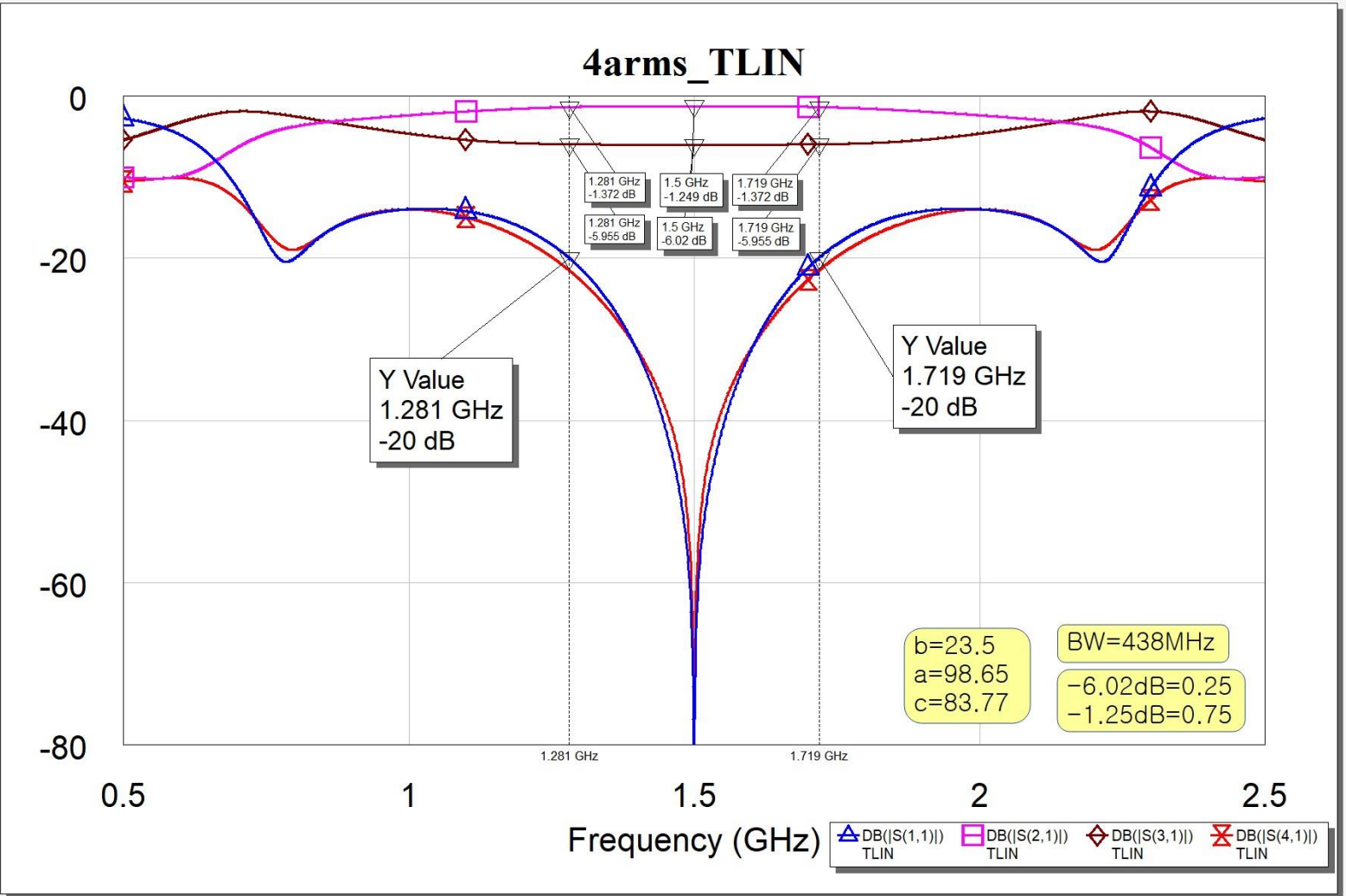
1단 대역폭 : 256MHz

2단 TLIN



2단 대역폭 : 464MHz

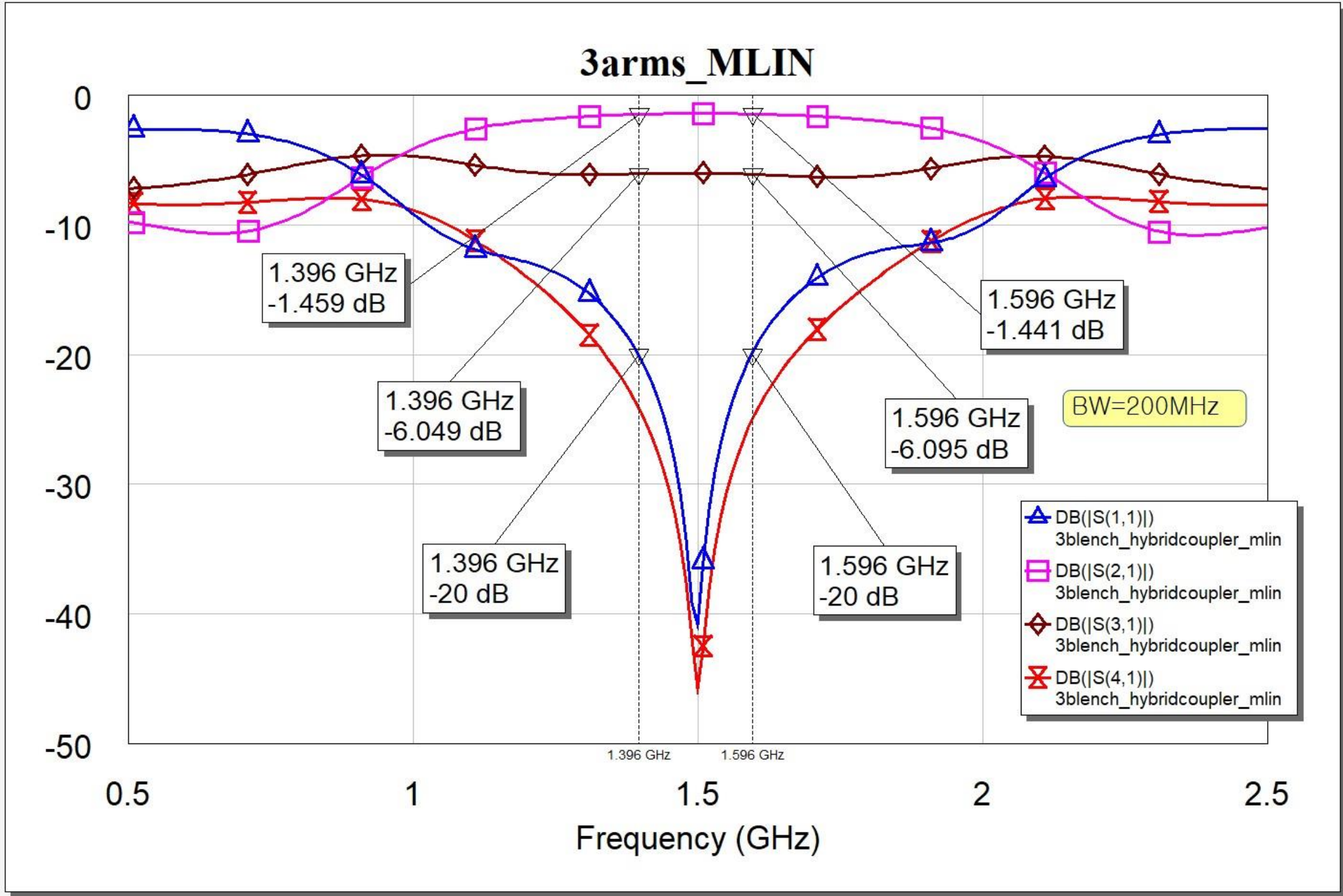
3단 50ohm TLIN



3단 대역폭 : 438MHz

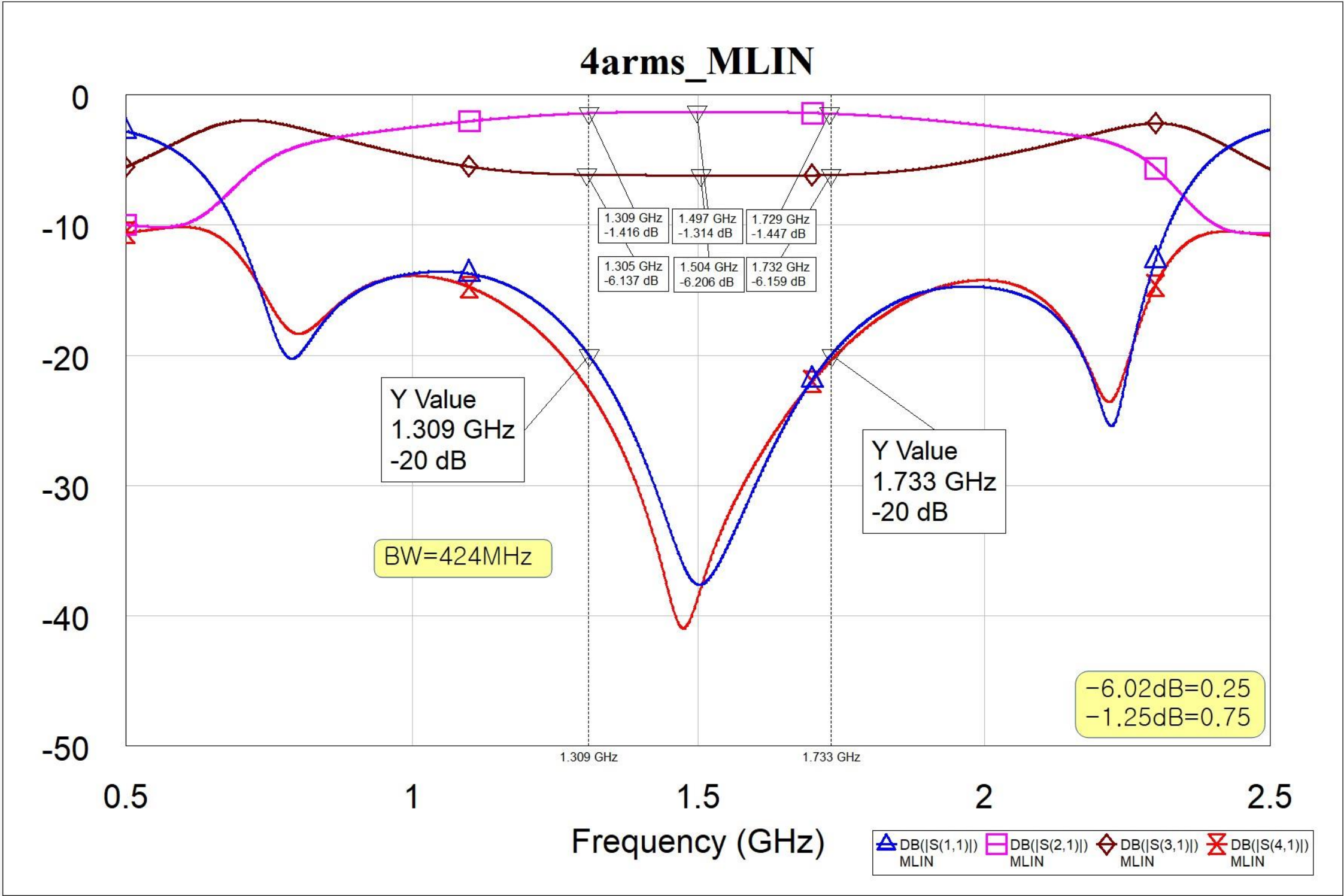
결과 및 분석

2단 MLIN



2단 대역폭 : 200MHz

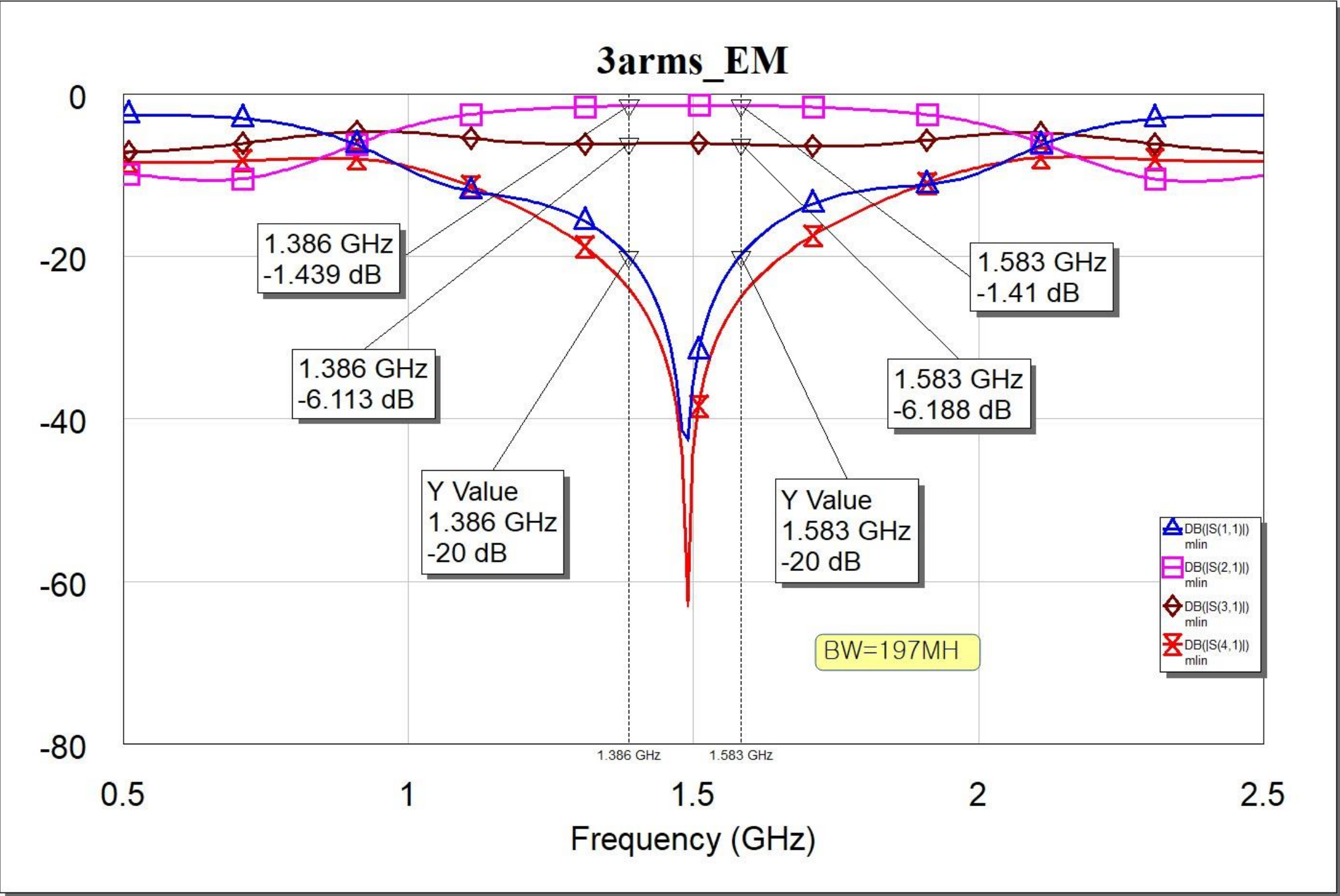
3단 MLIN



3단 대역폭 : 424MHz

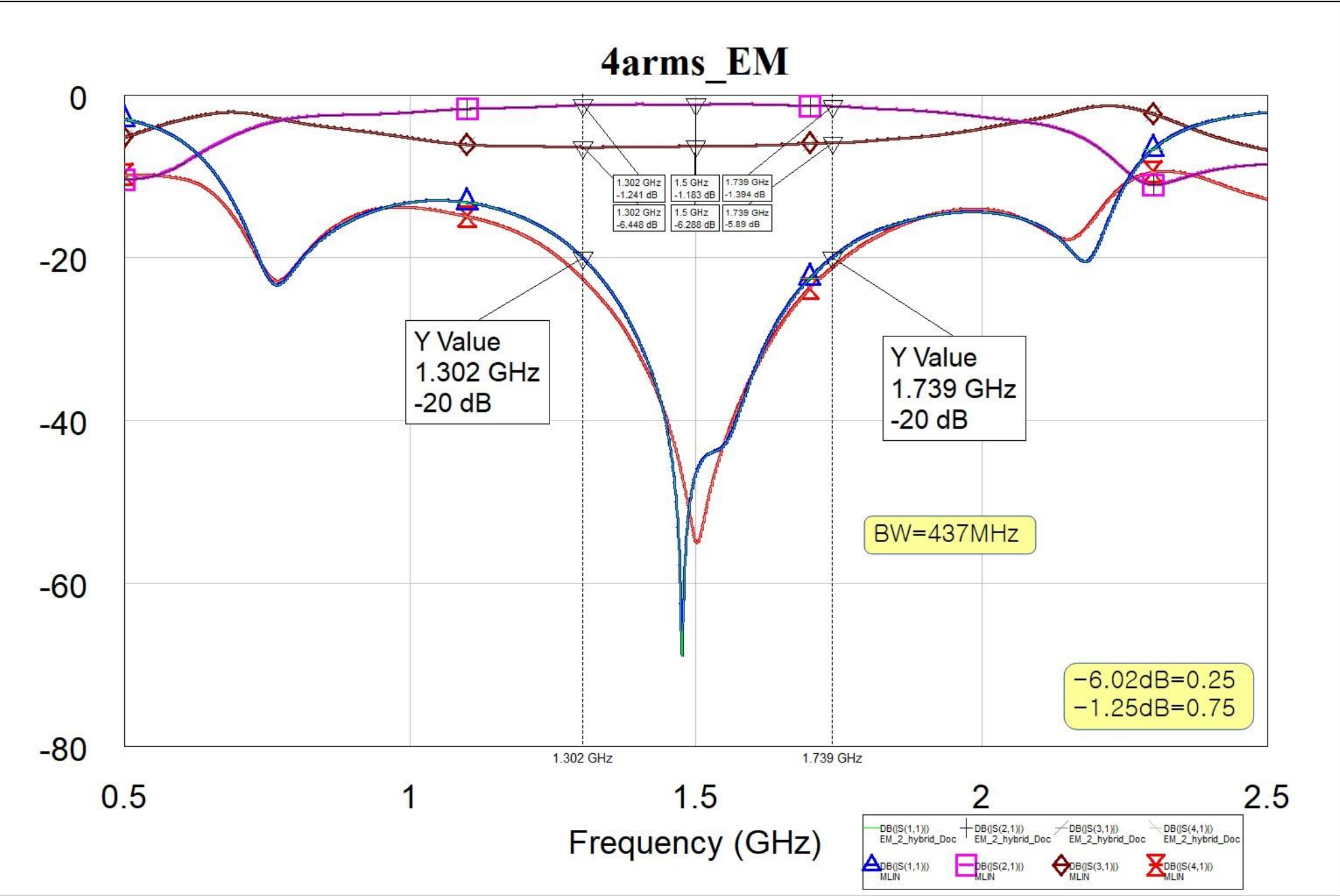
결과 및 분석

2단 EM



2단 대역폭 : 197MHz

3단 EM



3단 대역폭 : 437MHz

결과 및 분석

	1단	2단	3단
TLIN	256MHz	464MHz	500MHz
MLIN		200MHz	424MHz
EM		197MHz	437MHz

TLIN 트레이스 두께 고려 x

$\lambda/4$ 파장 변환기 적용

$\lambda/4$ 파장 변환기 적용

결과 및 분석

$S_{11}, S_{41} < -20\text{dB}$

S_{21}, S_{31} 전력분배 오차 10%이내

Phaze error < 0.1 조건에서

단수가 늘어남에 따라 대역폭이 증가하는 경향을 확인